

Auction-based solution for the ordering problem in robotic self-assembly

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Agenda

- Introduction
- Ordering in self-assembly processes description
- Auction-based assignment
- Results

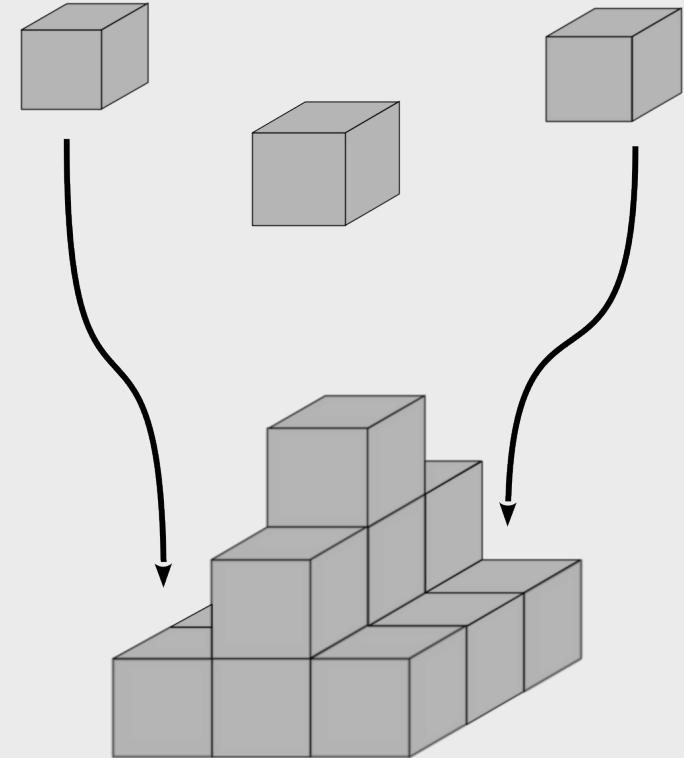


Introduction

Self-assembly is the problem where parts reorganize in space to build a desired structure.

Two areas are related to this problem:

- Ordering
 - How to establish an order between equal parts.
- Sequencing
 - Sequence of which the positions in the structure will be filled.



Problem Description

In **ordering** problems, the next robot to be used during construction may be selected from a group of equally capable robots.

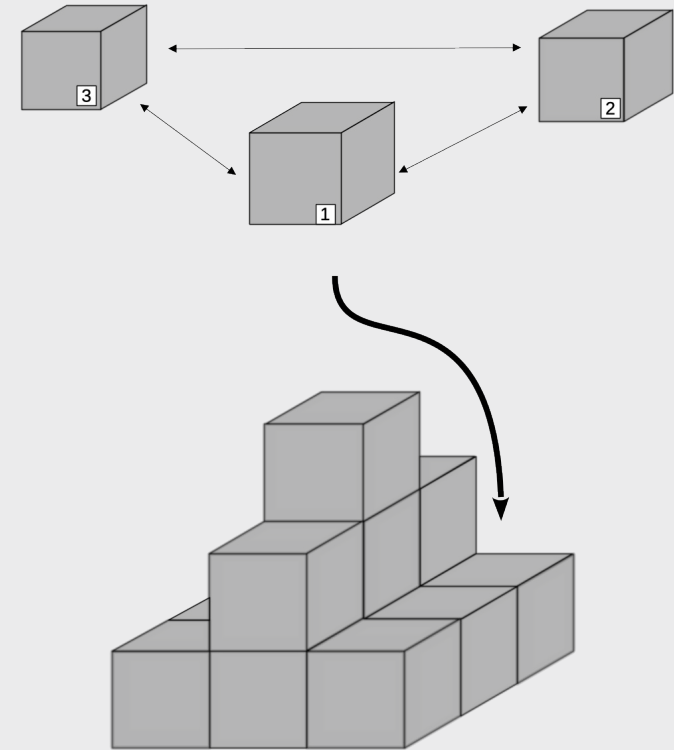
This is particularly relevant if only one part at a time should be added to the structure.

Ordering can be modelled as a **task assignment** problem.

$$\max \sum_{i=1}^{N_R} \left(\sum_{j=1}^{N_t} c_{ij} x_{ij} \right) \quad (1)$$

subject to

$$\begin{aligned} \sum_{j=1}^{N_t} x_{ij} &= 1 \quad \forall i \in \mathcal{I} \\ \sum_{i=1}^{N_R} x_{ij} &= 1 \quad \forall j \in \mathcal{J} \\ \sum_{i=1}^{N_R} \sum_{j=1}^{N_t} x_{ij} &= N_t = N_R \\ x_{ij} &\in \{0, 1\} \quad \forall (i, j) \in \mathcal{I} \times \mathcal{J} \end{aligned} \quad (2)$$



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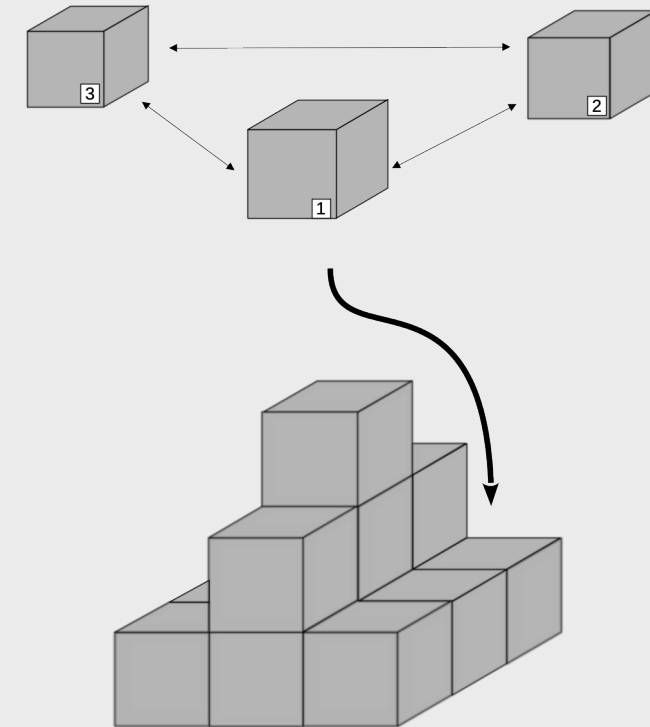
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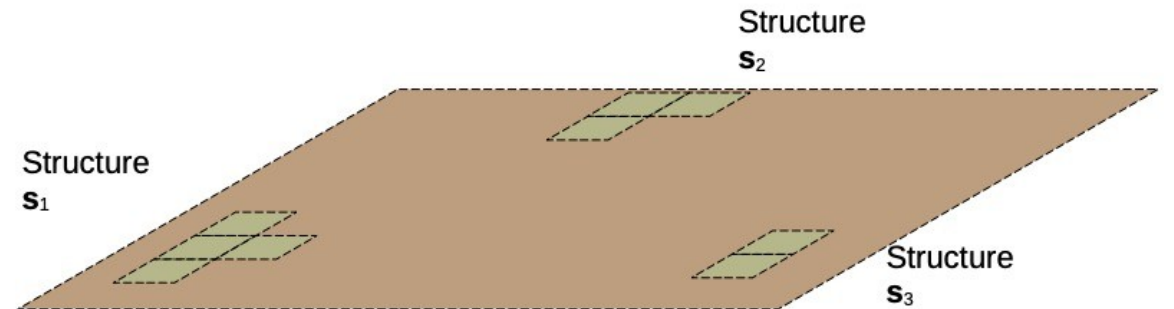
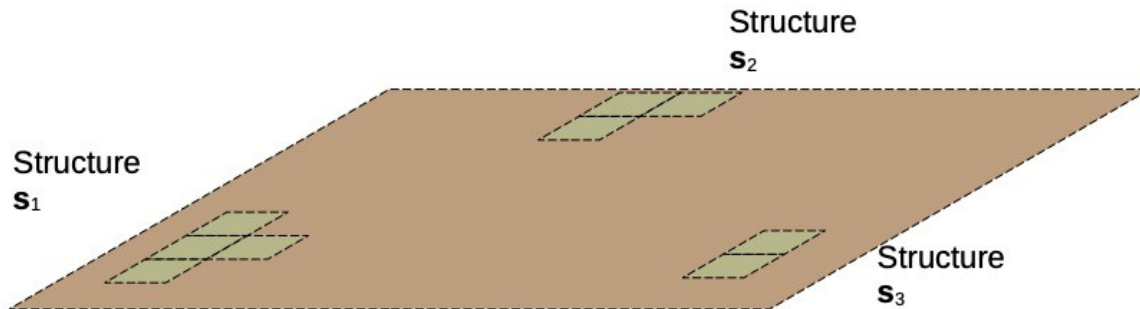
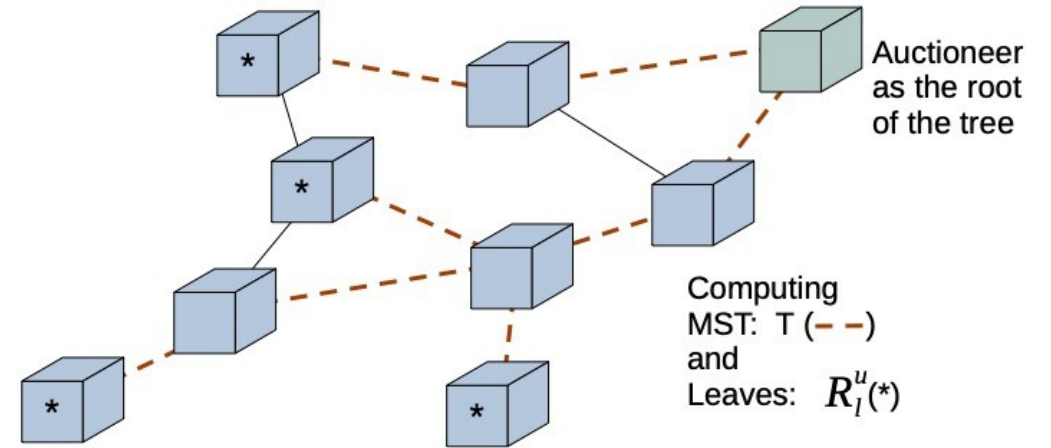
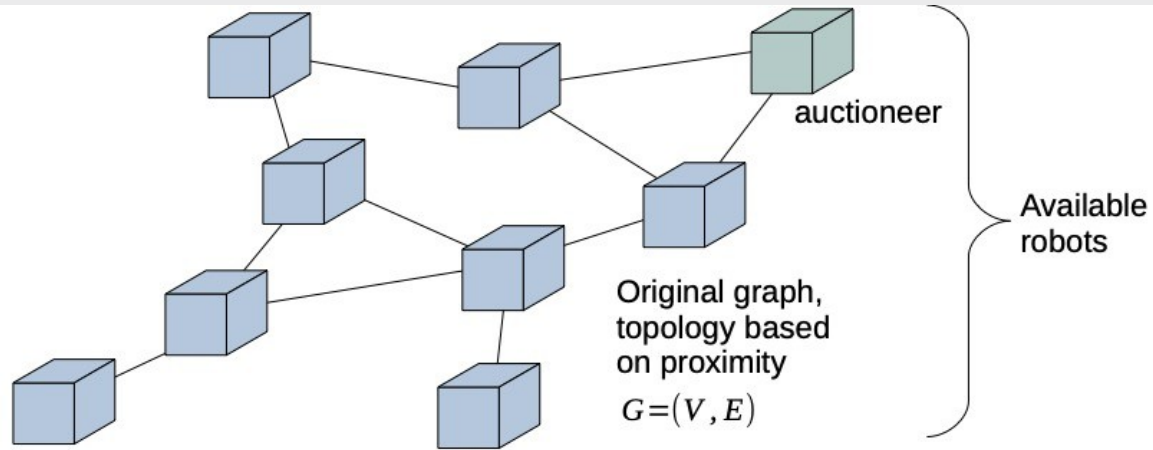
Ordering can be modelled as a **task assignment** problem.

Objective:

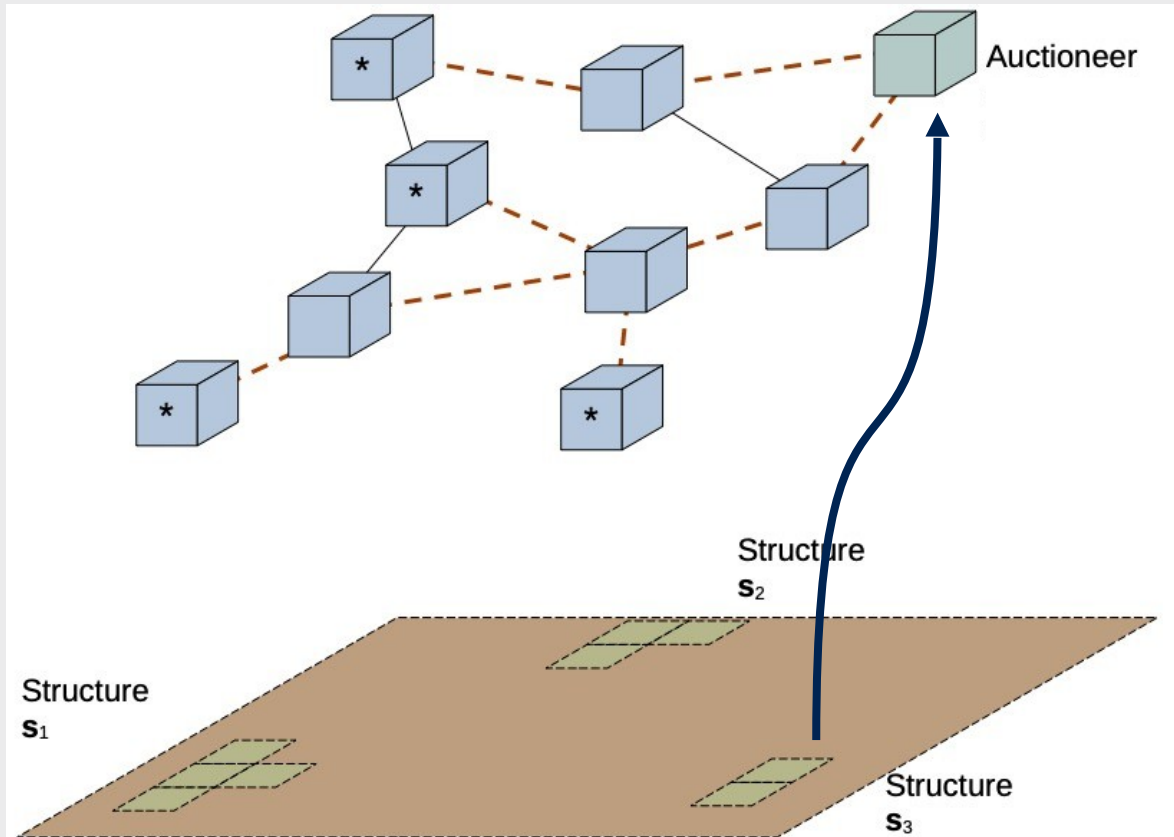
Determine the best robot in the group to be used as the next part in the construction process.



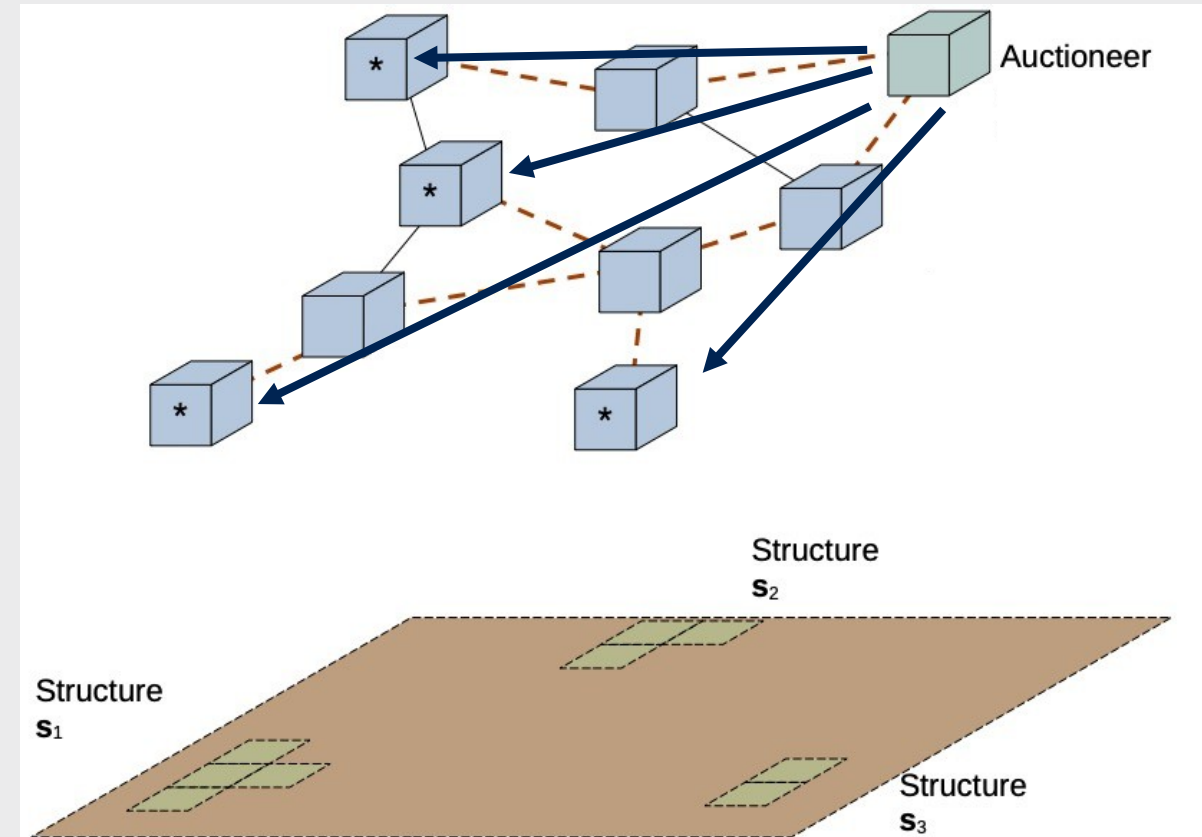
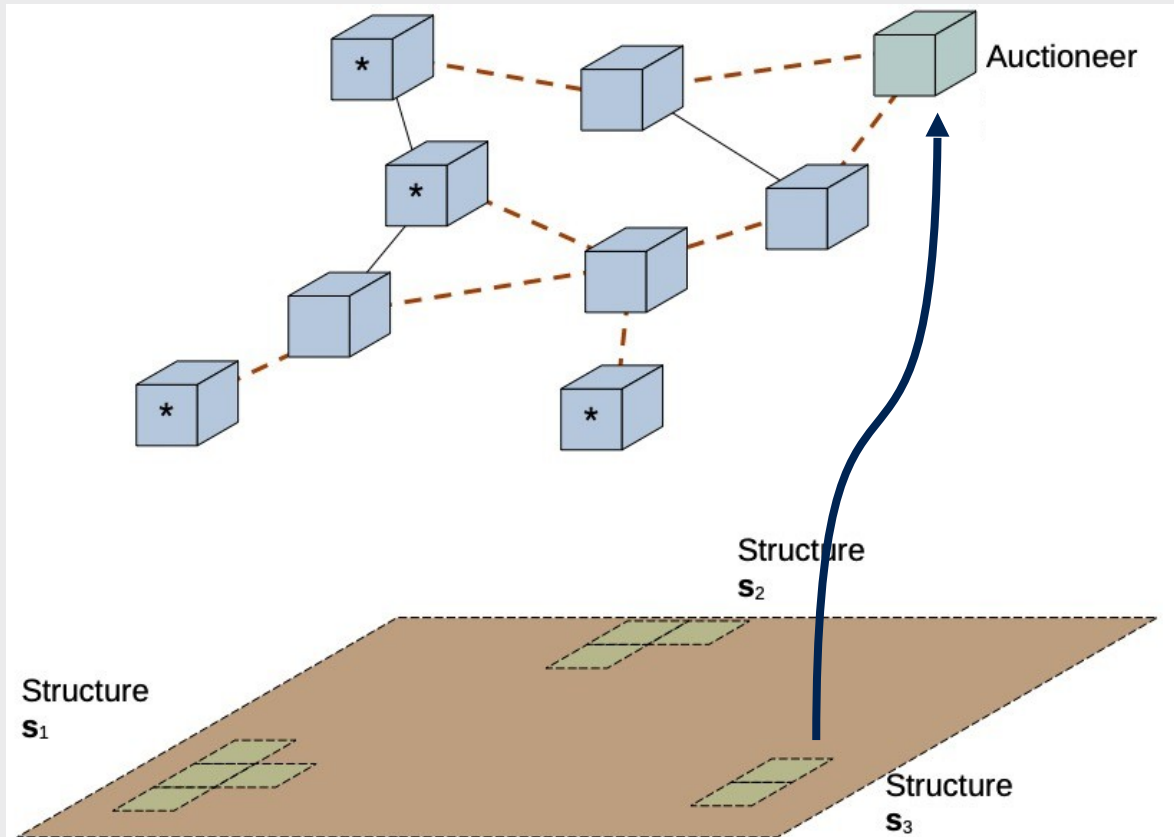
Auction-based assignment



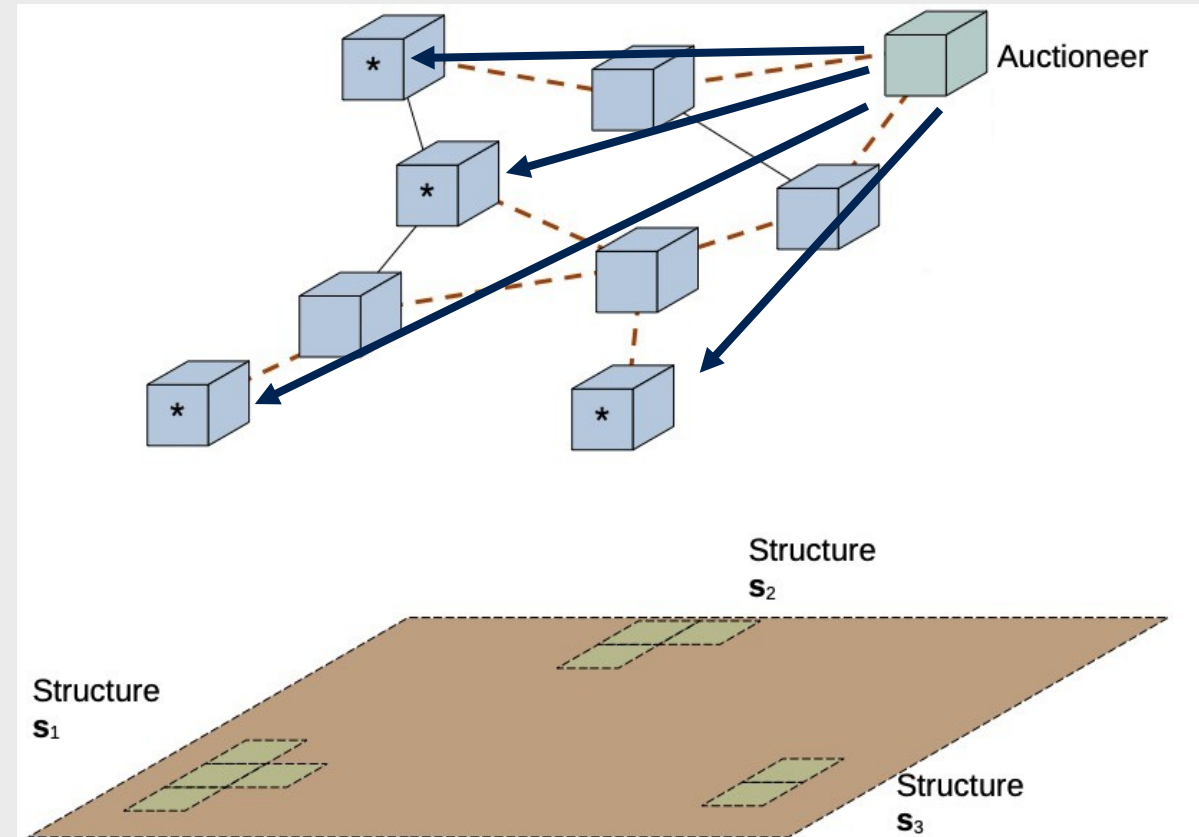
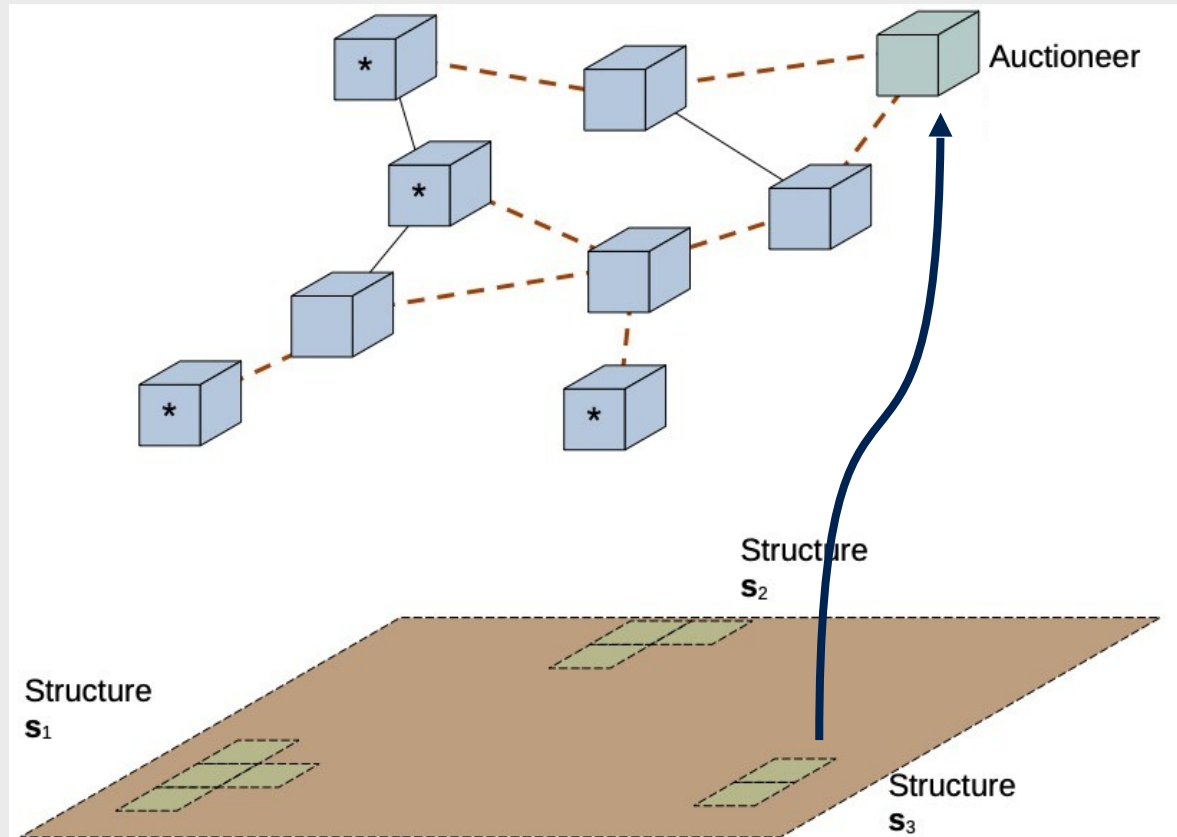
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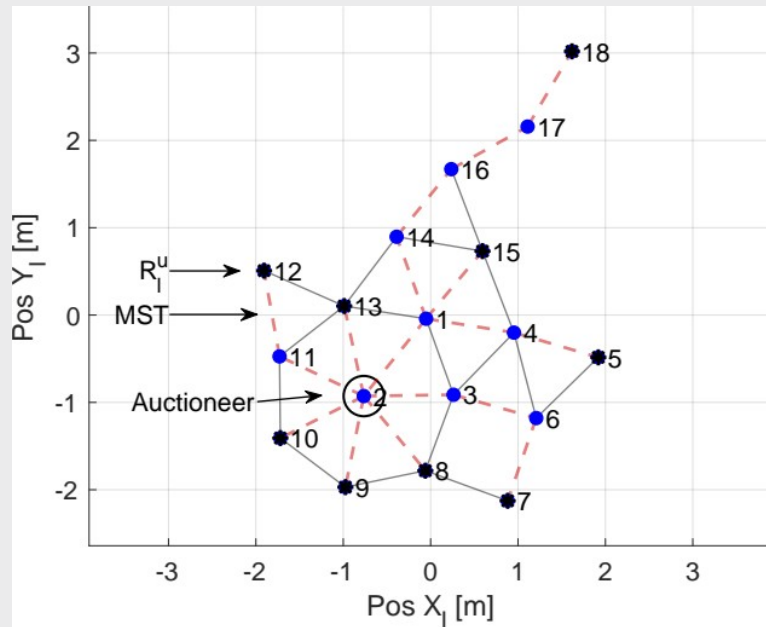


$$c_{ij}(i, p^s) = \frac{1}{b_i} + \frac{1}{||p_i - p^s||}$$

$$i^* = \operatorname{argmax}_i c_{ij}(i, p^s)$$

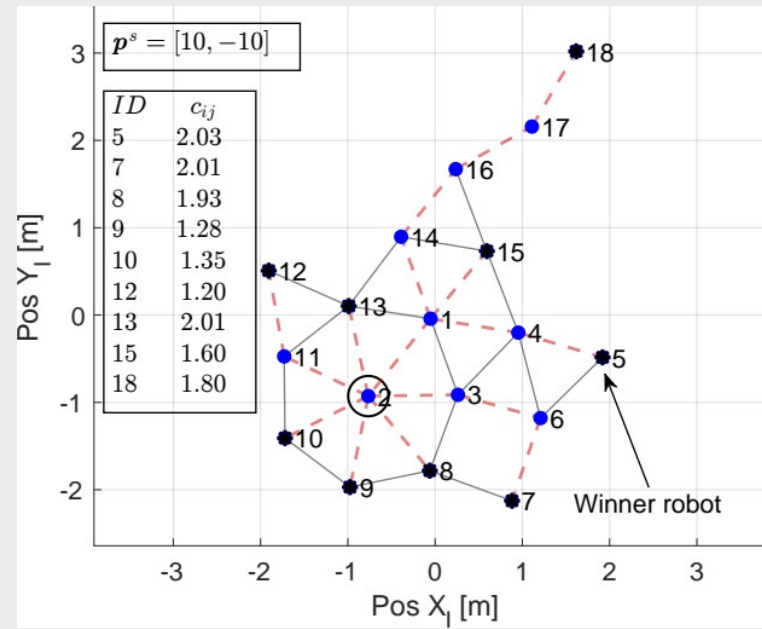
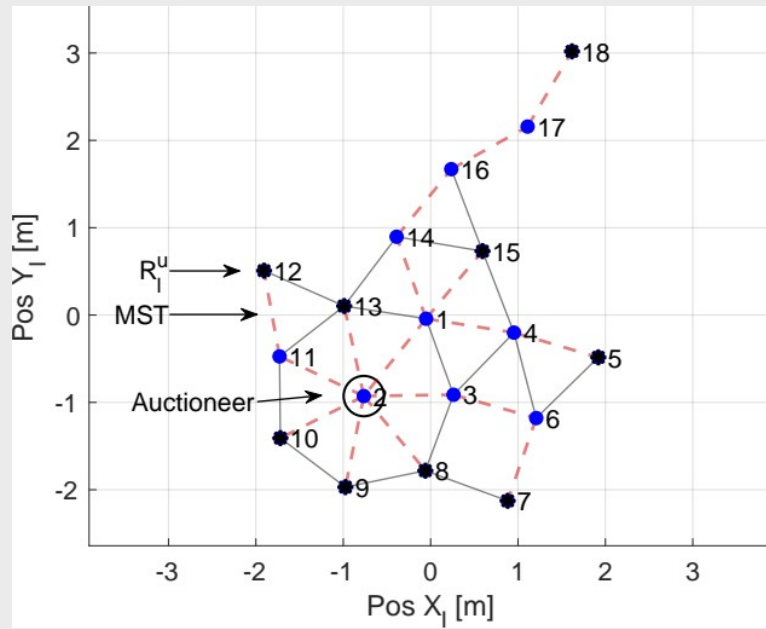
Results

Step by step



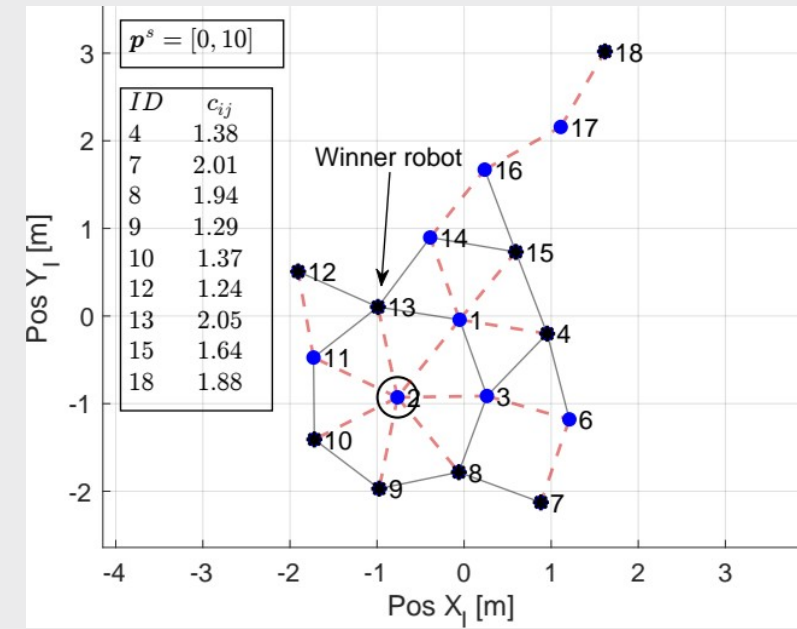
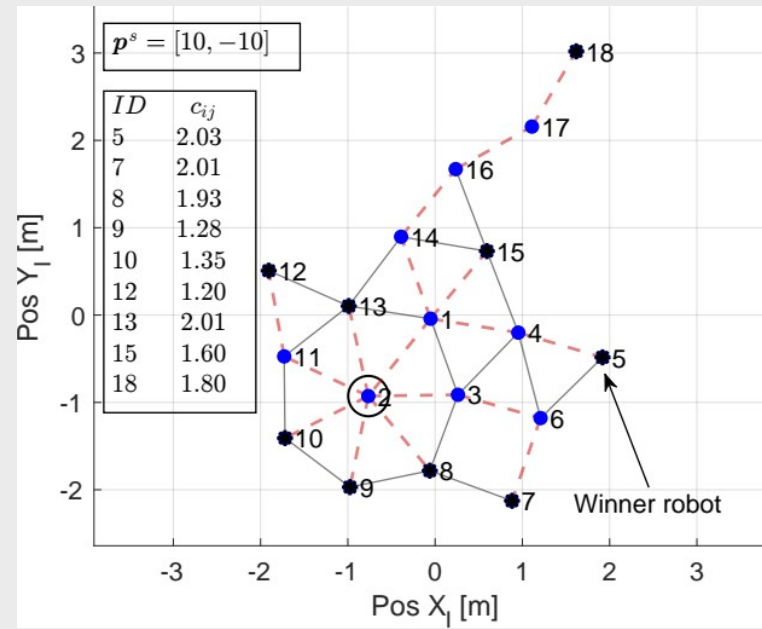
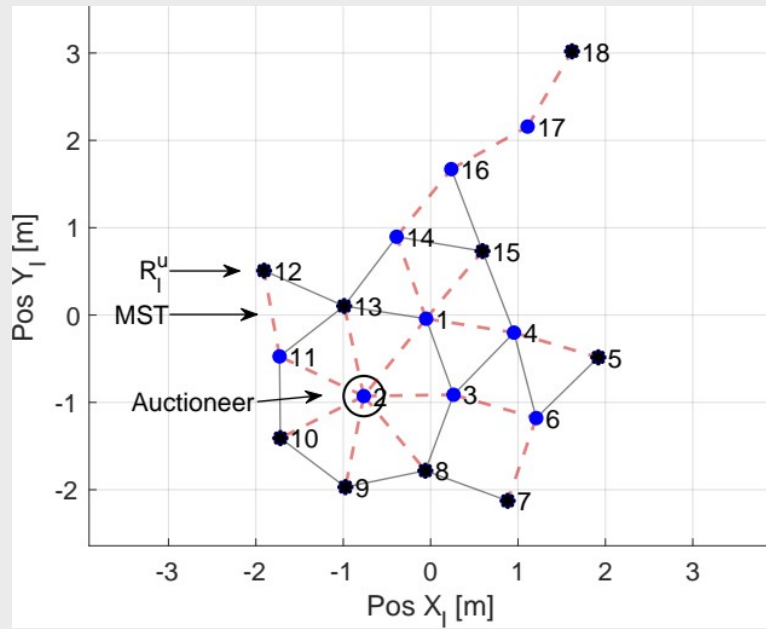
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Scenarios	Number of robots per structure	N_t
1	5, 6, 7	18
2	50, 60, 70	180
3	100, 100, 100	300



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$$b_r = \frac{1}{N_R} \sum_{i=1}^{N_R} \Delta b_{i,r},$$

$$\bar{b} = \frac{1}{N_r} \sum_{r=1}^{N_r} b_r,$$

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The assignment was performed with the computation of the MST and the identification of the leaf nodes, but the winner could be selected:

- Randomly
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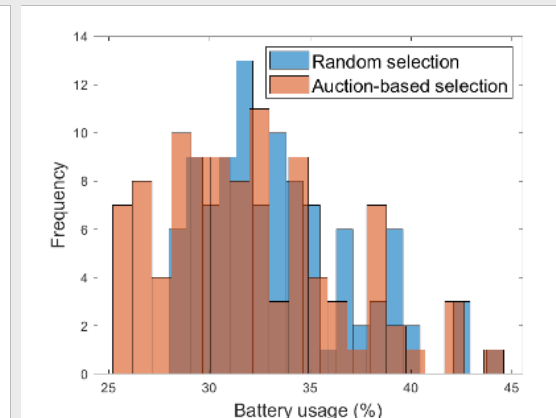
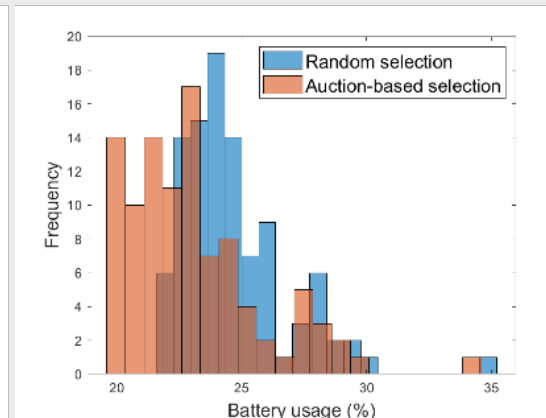
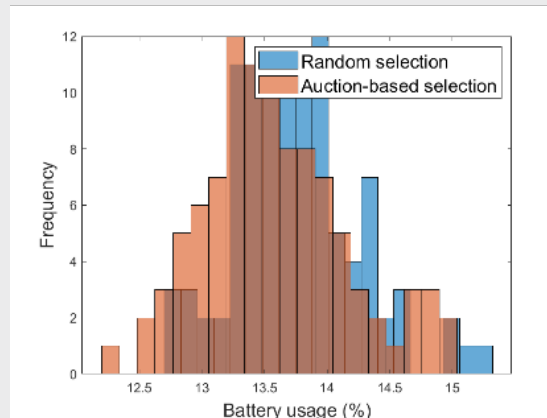
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Conclusion

- This work discussed how to compute **an order among equally capable robotic parts** such that they can be used in the as construction blocks in self-assembly tasks.
- The assignment was modelled as an **integer program** with binary decision variables (single assignment).
- The auction-based solution is proposed that uses a Minimum Spanning Tree **to identify the available robots** in the group.
- A communication network is **guaranteed** throughout the process.
- The proposed method is **more energy efficient** than a random selection, leading the robots to save power while waiting to be assigned.



Thank you

